

April 22, 2026

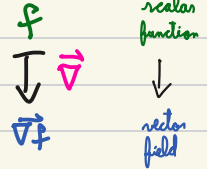
Recall: del operator: $\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$.

This ∇ operator acts on function on $\mathbb{R}^3$ (real or $\mathbb{R}^3$ vector valued):

• If $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ is a differentiable scalar function, then

∇ acts on f on the gradient:

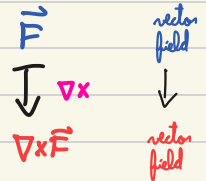
$$\vec{\nabla} f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = \vec{\nabla} f$$



• If $\vec{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a differentiable vector field, then

$\nabla \times$ acts on $\vec{F}$ on the curl:

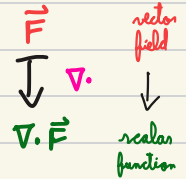
$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$



• If $\vec{F}: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is a differentiable vector field, then

$\nabla \cdot$ acts on $\vec{F}$ on the divergence:

$$\nabla \cdot \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$



Proposition (some properties of $\vec{\nabla}$, $\nabla \times$, $\nabla \cdot$): Suppose $c \in \mathbb{R}$ is a constant,

$f, g: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ are differentiable function, and

$\vec{F}, \vec{G}: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ are differentiable vector fields, then:

- 1) (linearity):
 - $\vec{\nabla}(f+g) = \vec{\nabla}f + \vec{\nabla}g$ and $\vec{\nabla}(cg) = c\vec{\nabla}g$
 - $\nabla \times (\vec{F} + \vec{G}) = \nabla \times \vec{F} + \nabla \times \vec{G}$ and $\nabla \times (c\vec{F}) = c\nabla \times \vec{F}$
 - $\nabla \cdot (\vec{F} + \vec{G}) = \nabla \cdot \vec{F} + \nabla \cdot \vec{G}$ and $\nabla \cdot (c\vec{F}) = c\nabla \cdot \vec{F}$

Leibniz rule

- 2) (product rule):
 - $\vec{\nabla}(fg) = (\vec{\nabla}f)g + f\vec{\nabla}g$

↓

These are very useful identities, but will not be crucial in this class

- $\nabla \times (f\vec{G}) = (\vec{\nabla}f) \times \vec{G} + f\nabla \times \vec{G}$
- $\nabla \cdot (f\vec{G}) = (\vec{\nabla}f) \cdot \vec{G} + f\nabla \cdot \vec{G}$
- $\nabla \cdot (\vec{F} \times \vec{G}) = (\nabla \times \vec{F}) \cdot \vec{G} - \vec{F} \cdot (\nabla \times \vec{G})$

Theorem ("curl of gradient"): Let $f: X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}$ be a C^2 function (all 1st and 2nd order partial derivatives exist and are continuous), then $\text{curl}(\nabla f) = \vec{0}$, i.e. $\nabla \times \nabla = \vec{0}$.

Theorem ("divergence of curl"): Let $\vec{F}: X \subseteq \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a C^2 vector field (all 1st and 2nd order partial derivatives exist and are continuous), then $\text{div}(\text{curl } \vec{F}) = 0$, i.e. $\nabla \cdot \nabla \times = 0$.

Application: We can use **curl** to rule out when a vector field in $\mathbb{R}^3$ (or $\mathbb{R}^2$) is conservative.

more generally ...

Proposition: Suppose $\vec{F}: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a C^1 vector field $\vec{F}(x_1, \dots, x_n) = (F_1(x_1, \dots, x_n), F_2(x_1, \dots, x_n), \dots, F_n)$. If $\vec{F}$ were a conservative vector field, then for any $i, j = 1, \dots, n$ we would have $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$.

remark: This can tell us when a vector field is not conservative, it can't tell when a vector field is conservative.

The proposition is a consequence of the equality of mixed 2nd order partial derivatives:

$f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ C^2 scalar function (potential for $\nabla \cdot f$)
 $\vec{F} = \nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) = (F_1, \dots, F_n)$
 then $f_{x_1 x_2} = \frac{\partial F_1}{\partial x_2} = \frac{\partial F_2}{\partial x_1} = f_{x_2 x_1}$, etc.

e.g. Find a potential function for each of the following v.f.
or indicate why they are not conservative

a) $\vec{F}(x, y, z) = 2x\hat{i} + 2y\hat{j} - 3z\hat{k}$ on $\mathbb{R}^3$

$$\int F_1 dx = \int 2x dx = x^2 + C_1(y, z)$$

$$\int F_2 dy = \int 2y dy = y^2 + C_2(x, z)$$

$$\int F_3 dz = \int -3z dz = -\frac{3}{2}z^2 + C_3(x, y)$$

$$f(x, y, z) = x^2 + y^2 - \frac{3}{2}z^2 \quad \vec{\nabla} f = \vec{F}$$

is conservative

b) $\vec{G}(x, y) = x\hat{i} + x\hat{j}$

$$\frac{\partial G_1}{\partial y} = 0 \neq \frac{\partial G_2}{\partial x} = 1$$

not conservative

c) $\vec{F}(x, y) = (-y, x)$

$$\frac{\partial F_1}{\partial y} = -1 \neq \frac{\partial F_2}{\partial x} = +1$$

not conservative

d) $\vec{G}(x, y, z) = \left(\frac{2x}{x^2+y^2+z^2}, \frac{2y}{x^2+y^2+z^2}, \frac{2z}{x^2+y^2+z^2} \right)$ on $\mathbb{R}^3 \setminus \{0\}$

$$\frac{\partial G_1}{\partial y} = \frac{-2x}{(x^2+y^2+z^2)^2} \cdot 2y = \frac{\partial G_2}{\partial x} = \frac{(-2y)(2x)}{(x^2+y^2+z^2)^2}$$

$$\int \frac{2x}{x^2+y^2+z^2} dx \stackrel{u=x^2+y^2+z^2}{=} \int \frac{1}{u} du = \ln|u| = \ln|x^2+y^2+z^2|$$

$$f(x, y, z) = \ln(x^2+y^2+z^2)$$

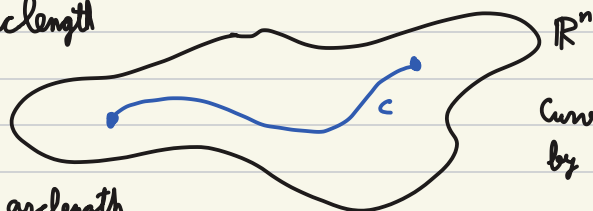
conservative

$$\vec{\nabla} f = \left(\frac{2x}{x^2+y^2+z^2}, \frac{2y}{x^2+y^2+z^2}, \frac{2z}{x^2+y^2+z^2} \right)$$

Line integrals (also called path integrals)

Suppose we have a curve C in $\mathbb{R}^n$ (parametrized by $\vec{x}: I \rightarrow C \subseteq \mathbb{R}^n$) and a function with domain C (or a region X in $\mathbb{R}^n$ containing C) which we want to integrate.

e.g. Arc length



Curve C parametrized by $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$

$$s = \int_a^b \|\vec{x}'(t)\| dt = \int_C 1 ds$$

ideally, the length of the curve should not depend on the parametrization
Does it depend? ~~It~~ could depend (mostly can change if you trace back over the path that has been walked over)

We will later need to keep this in mind when we differentiate between a general path integral and a path integral over a curve.
in these cases we want to "only trace the curve once"

Scalar line integrals and vector line integrals

Typical example is arc length. Idea in terms of Riemann sums:

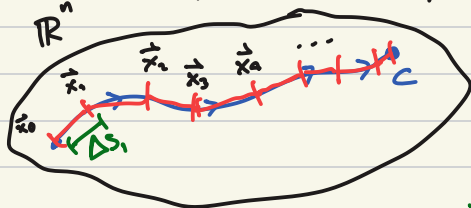
$$f: X \subseteq \mathbb{R}^n \rightarrow \mathbb{R} \quad \text{and}$$

$$\vec{x}: I \rightarrow X$$

we want the integral to be the limit of

$$\sum_{i=1}^n f(\vec{x}_i^*) \Delta s_i$$

↑
small arc length



notation: we will use s to denote the arclength ("distance traveled") along a parametric path and ds is an infinitesimal displacement along the parametrized path

scalar line integral

$$\int_{\vec{x}} f ds = \int_a^b f(\vec{x}(t)) \|\vec{x}'(t)\| dt$$

$$* ds = \|\vec{x}'(t)\| dt$$

for a parametric path $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$
and a scalar function f on $\Sigma \subseteq \mathbb{R}^n$

e.g. $\vec{x}(t) = (\cos(4t), \sin(4t), 3t)$ $0 \leq t \leq 2\pi$
and $f(x, y, z) = 3x + xy + z^3$

$$\int_{\vec{x}} f ds = ?$$

$$\vec{x}'(t) = (-4 \sin(4t), 4 \cos(4t), 3)$$

$$\|\vec{x}'(t)\| = \sqrt{4^2 \sin^2(4t) + 4^2 \cos^2(4t) + 3^2}^{1/2}$$

$$= \sqrt{25} = 5$$

$$f(\vec{x}(t)) = 3(\cos(4t)) + (\cos(4t))(\sin(4t)) + (3t)^3$$

$$= \cos(4t) [3 + \sin(4t)] + 27t^3$$

$$\int_0^{2\pi} f(\vec{x}(t)) \|\vec{x}'(t)\| dt = \int_0^{2\pi} (\cos(4t) [3 + \sin(4t)] + 27t^3) 5 dt$$

$$= \int_0^{2\pi} 15 \cos(4t) dt + \int_0^{2\pi} 5 \cos(4t) \sin(4t) dt$$

$$+ 5 \cdot 27 \int_0^{2\pi} t^3 dt$$

$$= \frac{15}{4} \sin(4t) + \frac{5}{8} \sin^2(4t) + \frac{5 \cdot 27}{4} t^4 \Big|_{t=0}^{t=2\pi}$$

$$= \frac{5 \cdot 27}{4} (2\pi)^4 = 5 \cdot 27 \cdot 4\pi^4$$